

COS 433 Problem Set 0

September 2, 2026

Due date: Wednesday, 9/9/2026 at 11:59pm.

See [the course website](#) for our submission and grading policies.

Problem 1: Survey and check-in sign-up

(a) Briefly describe your background. What is your major? What experience do you have with the following?

- Algorithms and analysis of algorithms
- Number theory
- Probability

(b) In a few words, why are you interested in taking this course?

(c) Sign up for Check-in 1 at this calendar link: <https://calendar.app.google/vKbRnPgXaJz14BvU6>.

Problem 2: Statistical Distance (Probability Review)

Let X_0 and X_1 denote random variables over a finite outcome set $\mathcal{X}$. Define the *statistical distance* between X_0 and X_1 as

$$\Delta(X_0, X_1) = \max_{T \subset \mathcal{X}} \left| \Pr[X_0 \in T] - \Pr[X_1 \in T] \right|.$$

That is, $\Delta(X_0, X_1)$ is the largest *distinguishing advantage* that can be attained by some “test” event $T \subset \mathcal{X}$ applied to X_0 and X_1 .

(a) Prove that Δ satisfies the triangle inequality:

$$\Delta(X, Z) \leq \Delta(X, Y) + \Delta(Y, Z).$$

(b) Prove that

$$\Delta(X_0, X_1) = \frac{1}{2} \sum_{x \in \mathcal{X}} \left| \Pr[X_0 = x] - \Pr[X_1 = x] \right|.$$

Problem 3: Perfect Secrecy.

Let (Enc, Dec) be a secret-key encryption scheme with message space $\mathcal{M}$.

(a) Suppose that M is uniformly distributed over two distinct messages m_0, m_1 . Prove that Eve's optimal probability of guessing M from C is exactly

$$\frac{1}{2} + \frac{1}{2} \cdot \Delta(\text{Enc}(K, m_0), \text{Enc}(K, m_1)).$$

This says that $\Delta(C_0, C_1)$ (for $C_b = \text{Enc}(K, m_b)$) characterizes how far a scheme is from satisfying perfect secrecy.

(b) Suppose that the encryption scheme satisfies *perfect secrecy* for uniformly random messages. Recall that in our definition from class, the key k was assumed to be sampled uniformly at random.

But what if we use imperfect randomness, and so the secret key is not uniformly random? Let $\tilde{K}$ be a random variable satisfying $\Delta(\tilde{K}, K) \leq \epsilon$, where the random variable K is distributed uniformly.

Prove that for a uniformly random message M , and all functions Eve,

$$\Pr[\text{Eve}(\text{Enc}(\tilde{K}, M)) = M] \leq \frac{1}{|\mathcal{M}|} + \epsilon.$$

Relatedly, prove that for all pairs of messages m_0, m_1

$$\Delta(\text{Enc}(\tilde{K}, m_0), \text{Enc}(\tilde{K}, m_1)) \leq 2\epsilon.$$

Problem 4: Tail Inequalities (Probability Review)

In the following problem, you may assume for simplicity that all of the relevant random variables take values in some finite outcome space.

(a) **Markov's Inequality.** Let X be a non-negative real random variable. Prove that for any $t > 0$,

$$\Pr[X > t] \leq \frac{\mathbb{E}[X]}{t}.$$

(b) **Chebyshev's Inequality.** Let X be a real random variable with variance σ^2 . Prove that for any $t > 0$,

$$\Pr[|X - \mathbb{E}[X]| > t] \leq \frac{\sigma^2}{t^2}.$$

(Hint: Apply Markov's Inequality to a function of X)

(c) **Averaging Independent Random Variables.** Let $X_1, \dots, X_n$ be a collection of n *independent* random variables. Suppose that for some constant σ , we are guaranteed that $\text{Var}[X_i] \leq \sigma^2$ for every i .

Let $X = \frac{1}{n} \sum_i X_i$ be the average of $X_1, \dots, X_n$. Prove that

$$\Pr[|X - \mathbb{E}[X]| > t] \leq \frac{\sigma^2}{n \cdot t^2}.$$

Why is it important to assume that the X_i are independent? Can you state a relaxation of the independence assumption on $X_1, \dots, X_n$ that still allows us to deduce the inequality?

(d) Testing the bias of a coin. Suppose that you have a coin that is either unbiased (heads/tails occur with probability $1/2$ each when you flip the coin) or ϵ -biased (heads occurs with probability $1/2 + \epsilon$). How many times, as a function of ϵ , do you need to flip the coin in order to correctly guess which is the case with 99% accuracy?